

Eigenvectors and eigenvalues

Lecture notes for myself

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May 20, 2026

1 Introduction

Some linear transformations (LTs) have unique vectors, which simply scale when the transformation is applied to them:

$$T(\vec{v}) = \lambda \vec{v}, \quad \lambda \in \mathbb{R}. \quad (1)$$

In matrix form, if the matrix $\mathbf{A}$ represents T in the standard basis set, then

$$\mathbf{A}\vec{v} = \lambda \vec{v}. \quad (2)$$

(where λ is the same scalar as in Equation 1)

We call such vectors the **Eigenvectors** of $\mathbf{A}$ with the corresponding scale factors λ their respective **Eigenvalues**.

Example 1.1 A matrix and its eigen-vectors and -values

Let

$$A = \begin{bmatrix} 2 & 0 \\ -3 & -1 \end{bmatrix}.$$

The vectors $\vec{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ are both eigenvectors of $\mathbf{A}$: ...

